

Robotic Manipulation Coursework Report

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1 Task 1

1.1 Frame Assignment and Inverse Kinematics

To model the robot arm, DH convention was adopted. This provides a systematic way to assign coordinate frames to each joint and to derive the forward kinematics in a compact matrix form. The manipulator was treated as a 4-degree-of-freedom serial arm consisting of one base rotation joint and three arm joints.

The base frame $\{0\}$ was placed at the robot base. Joint 1 corresponds to rotation about the vertical axis, allowing the arm to sweep in the horizontal plane. Frames $\{1\}$ to $\{4\}$ were then attached successively along the links so that joints 2, 3 and 4 describe the shoulder, elbow and wrist motions. After the first joint, the remaining links move in a common vertical plane.

The first twist angle of 90° accounts for the change between the base yaw axis and the plane of the arm. The remaining twist angles are zero because links 2 to 4 are arranged as a planar chain. In addition, joint offsets were included in the implementation to align the mathematical zero angles with the actual motor zero positions of the robot. The corresponding DH parameter table and frame assignment figure are provided in the Appendix. This frame assignment was chosen because it reflects the physical structure of the manipulator closely and leads to a simple forward kinematic model. In particular, separating the base yaw from the planar arm motion makes the inverse kinematics easier to solve geometrically.

Rather than solving the full 4-DOF inverse kinematics problem in one step, the problem was decomposed into simpler parts. This was possible because the first joint only controls the azimuth of the arm, while the remaining joints determine the position and orientation of the end-effector in a vertical plane.

For a target position (x, y, z) , the base rotation angle was computed using $\text{atan2}()$, which aligns the arm with the target in the horizontal plane. After this, the problem reduces to the radial-height plane, where the horizontal distance from the base axis to the target can be found.

Since the end-effector has a non-zero length, the inverse kinematics was solved using the wrist centre rather than the tool tip directly. If the desired end-effector pitch is ϕ , the wrist centre coordinates are

$$r_w = r - L_4 \cos \phi \quad (1)$$

$$z_w = z - d_1 - L_4 \sin \phi \quad (2)$$

where L_4 is the final link length and d_1 is the base height offset. This reduces the problem to a standard 2-link planar inverse kinematics problem for joints 2 and 3.

Using the cosine rule, the elbow joint angle was found, which gives two possible solutions corresponding to the elbow-up and elbow-down configurations. For each valid θ_3 , the shoulder angle was then calculated. Finally, the wrist angle was obtained from the orientation constraint so that the combined rotation of joints 2, 3 and 4 produced the required tool pitch.

In practice, inverse kinematics may produce two valid solutions or no exact solution for a chosen wrist orientation. To improve robustness, the controller did not assume a single fixed ϕ . Instead, it searched through candidate wrist orientations and selected solutions that avoid collisions, with preference given to the elbow-up configuration to reduce the risk of contact with the table.

This inverse kinematics approach was chosen because it is computationally efficient, easy to interpret and debug, and well suited to the robot geometry. By separating the base rotation from the planar arm calculation, the problem becomes much simpler than a fully numerical inverse kinematics method. The geometric solution is also convenient for enforcing joint limits and for choosing between two configurations in a controlled way.

2 Task 3 - Robotic Calligraphy and Paper Placement

2.1 Task Motivation

Our task combines paper manipulation with robotic calligraphy: the robot autonomously writes Chinese characters on a sheet of paper and then attaches the finished artwork to a wall. This task integrates indirect grasping, object placement, end-effector tool exchange, and fine-motion control.

Writing and displaying calligraphy is a traditional activity during the Chinese Spring Festival, where people hang Spring Festival couplets (Chunlian) to express wishes for good fortune. Inspired by this tradition, our system demonstrates how a robot can write calligraphic characters and place the finished work on a wall.

The workflow consists of three stages. First, the robot retrieves and positions the paper using magnetic manipulation, where two magnets trap the paper to allow indirect lifting. Second, the robot exchanges the magnetic tool for a brush holder and executes a sequence of strokes to write the character. Finally, the robot retrieves the paper again and attaches it to a metal surface on the wall using the magnet.

2.2 Task Execution

The task consists of a complete manipulation and writing pipeline: the robot first uses a magnetic tool to pick up and place the paper on a working surface, then switches to a brush tool to execute programmed calligraphy strokes, and finally retrieves the paper again using magnetic manipulation to attach the finished artwork to a metal surface on the wall.

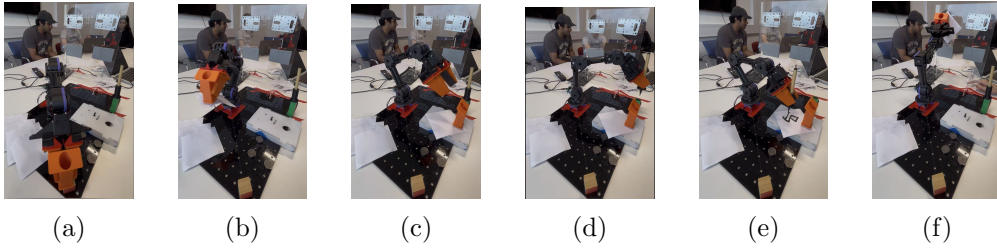


Figure 1: Execution sequence of the robotic manipulation and calligraphy process.

2.3 Mechanical Design

The task involves two distinct end-effectors: a magnetic tool for paper handling and a brush holder for writing.

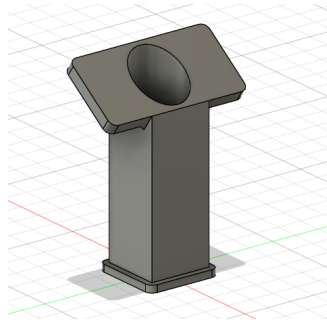


Figure 2: Magnetic tool

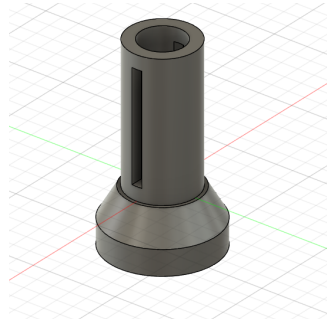


Figure 3: Brush tool

Magnetic Tool for Paper Manipulation: A tool with cylindrical cavity containing an internal magnet enables indirect grasping by attracting another magnet placed beneath the paper. This design creates a magnetic sandwich where the paper is trapped between two opposing magnetic forces. The magnet adheres more strongly to external metal surfaces than to the carried magnet through the

housing material. This property allows the robot to retrieve the tool while leaving the magnet with paper attached to the wall.

Brush Tool: A 3D-printed cylindrical sleeve fitted onto a pen to increase its diameter, enabling the robotic arm to achieve a more stable and reliable grip during manipulation.

2.4 Challenges

Paper Grasping and Transport: The primary challenge lies in achieving precise position control to ensure the paper, as a flexible object, can be reliably secured between the two magnets. When the paper is softer than in the video, it may sag under gravity and contact the table edge and surface, leading to deformation and task failure. Therefore, the motion path must be carefully planned to account for gravity, ensuring the paper remains properly supported and is placed flat on the workspace.

Brush Calligraphy Control: Writing calligraphy introduces additional control difficulties. Brush pressure must be regulated through precise vertical position control to maintain appropriate contact between the brush and the paper. Excessive pressure may cause ink spreading and distorted strokes, while insufficient pressure may lead to incomplete or faint lines. Stroke speed also influences ink delivery and line thickness, meaning that the robot must vary its velocity along different segments of the trajectory. Furthermore, correct pen-down and pen-up transitions with appropriate brush orientation are necessary to produce visually consistent strokes and maintain the aesthetic characteristics of calligraphy.

Tool Exchange: The robot must also switch between the magnetic tool and the brush holder during the task. This requires accurate positioning to ensure reliable grasping and release of the tools.

Overall, the task requires coordinated integration of indirect grasping, flexible object handling, precise trajectory control and reliable tool switching within a single robotic system.

2.5 Trajectory Refinement and Calligraphy Interface

To address the challenges of generating precise brush stroke trajectories, we developed an iterative approach combining kinesthetic teaching with interactive trajectory refinement:

Kinesthetic Teaching: The robot’s torque control is disabled, allowing a human operator to physically guide the robot’s end-effector while writing the desired character on paper. This method generates a rough initial trajectory that captures the overall stroke patterns and sequences.

Interactive Refinement UI: A custom PyGame interface was developed for fine trajectory adjustment. Each stroke is represented by discrete interpolation points that can be interactively selected and moved by the operator. This allows intuitive refinement of the trajectory, including vertical position (brush pressure), path curvature, temporal spacing between points (stroke speed), and brush orientation along the stroke.



Figure 4: PyGame-based user interface for trajectory refinement

Once refined trajectories are finalized in the UI, they are exported for autonomous robot execution. This hybrid teaching and refinement methodology bridges the gap between human intuition for calligraphic aesthetics and robotic precision for consistent execution.

3 Appendix

3.1 DH Parameters

Table 1 lists the DH parameters used for the 4-DOF manipulator model.

Table 1: DH parameters of the manipulator

Joint i	α_{i-1}	a_{i-1}	d_i	θ_i
1	90°	0	d_1	θ_1
2	0	L_2	0	θ_2
3	0	L_3	0	θ_3
4	0	L_4	0	θ_4

3.2 Frame Assignment

Figure 5 shows the coordinate frame assignment used for the manipulator model.

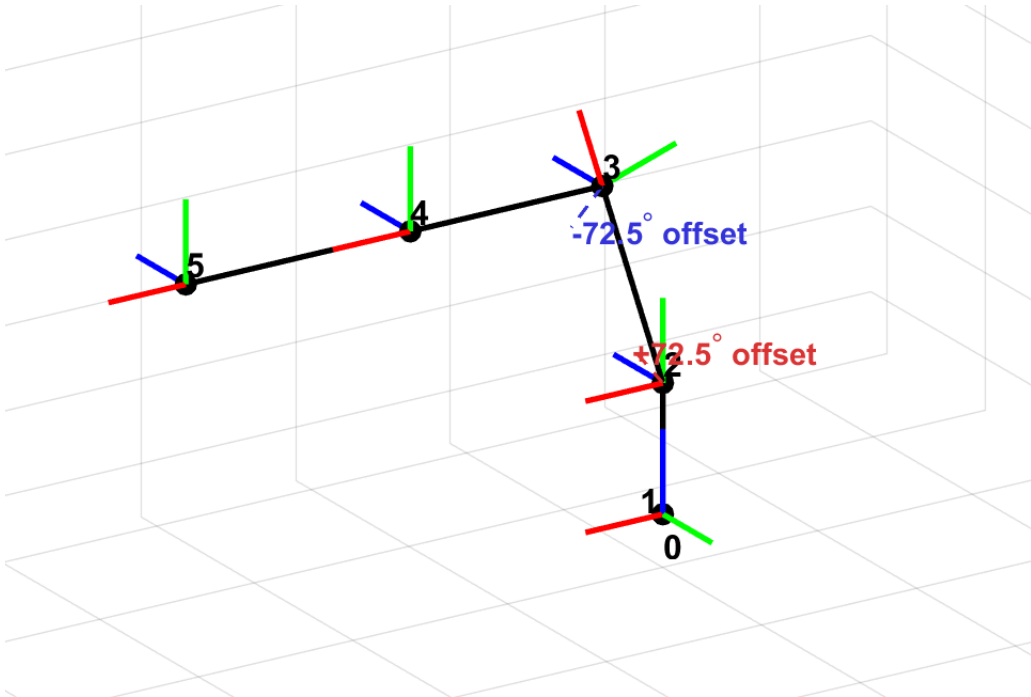


Figure 5: Coordinate frame assignment for the robot manipulator.

3.3 Inverse Kinematics Calculations

For a target position (x, y, z) , the base rotation angle was computed as

$$\theta_1 = \text{atan2}(y, x) \quad (3)$$

which aligns the arm with the target in the horizontal plane. After this, the spatial problem reduces to a planar problem in the radial-height plane, where

$$r = \sqrt{x^2 + y^2} \quad (4)$$

represents the horizontal distance from the base axis to the target.

Since the end-effector has a non-zero length, the inverse kinematics was solved using the wrist centre rather than the tool tip directly. If the desired end-effector pitch is ϕ , the wrist centre coordinates are

$$r_w = r - L_4 \cos \phi \quad (5)$$

$$z_w = z - d_1 - L_4 \sin \phi \quad (6)$$

where L_4 is the final link length and d_1 is the base height offset. This reduces the problem to a standard 2-link planar inverse kinematics problem for joints 2 and 3.

Using the cosine rule, the elbow joint angle was found from

$$\cos \theta_3 = \frac{r_w^2 + z_w^2 - L_2^2 - L_3^2}{2L_2L_3} \quad (7)$$

which gives two possible solutions corresponding to the elbow-up and elbow-down configurations. For each valid θ_3 , the shoulder angle was then calculated by

$$\theta_2 = \text{atan2}(z_w, r_w) - \text{atan2}(L_3 \sin \theta_3, L_2 + L_3 \cos \theta_3) \quad (8)$$

Finally, the wrist angle was obtained from the orientation constraint

$$\theta_4 = \phi - \theta_2 - \theta_3 \quad (9)$$

so that the combined rotation of joints 2, 3 and 4 produced the required tool pitch.

3.4 Task 3 Additional Hardware Components

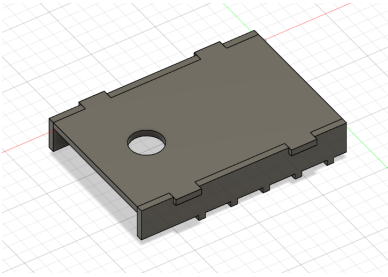


Figure 6: Table board used as the writing surface.

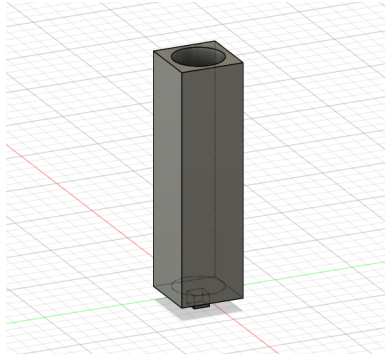


Figure 7: Pen base used to hold the calligraphy brush.

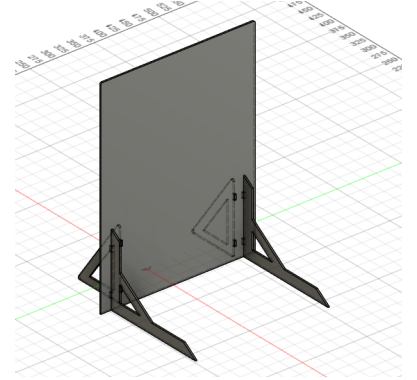


Figure 8: Display board used to attach the finished calligraphy.